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# Paleomagnetism of the Red Lake greenstone belt, northwestern Ontario: Possible evidence for the timing of gold mineralization

Vincenzo Costanzo-Alvarez [1] and David J. Dunlop

*Geophysics Laboratory, Department of Physics, University of Toronto, Toronto, Ont. M5S 1A7, Canada*

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ABSTRACT

We report paleomagnetic results from (1) largely unaltered batholiths and (2) sheared and altered units in the Archean Red Lake greenstone belt, a major gold-producing region in the Uchi subprovince of northwestern Ontario. Group 1 includes the Hammell Lake, Killala Baird, Trout Lake and Little Vermilion Lake batholiths, which were intruded between 2730 and 2700 Ma ago (U/Pb, zircon). Group 2 includes the Howey diorite, the Dome Stock, and dikes and andesites from the Dickenson mine. These units, from the mineralized deformation zones (DZs) between the batholiths, were sheared and hydrothermally altered in the same event that caused gold mineralization in supracrustal units. The batholiths carry a stable reversed remanence (RLR), with $D = 159°$, $I = -74°$ ($k = 20$, $\alpha_{95} = 16°$, $N = 4$ units, 14 sites). RLR has high coercivities, unblocking temperatures and remanence intensities and appears to be a primary thermal remanence dating from $\approx 2700$ Ma. The altered units from the gold-bearing DZs carry a different remanence (RLG), with $D = 180°$, $I = -57°$ ($k \approx 23$, $\alpha_{95} = 3°$, $N = 3$ units, 6 sites). RLG also has relatively high coercivities and unblocking temperatures but its magnetization intensities are 1–2 orders of magnitude lower than those of RLR. Opaque minerals and silicates observed in thin section are highly altered. RLG is therefore believed to be a chemical or thermochemical remanence acquired $\approx 2580$ Ma ago during the late stages of cooling and gold mineralization. Seeing as RLG characterizes the shear zone rocks but is absent from unaltered rocks outside the DZs, it could be used as a rapid prospecting tool in delineating areas of alteration and possible mineralization. The paleopoles for RLR ($\approx 2700$ Ma) and RLG ($\approx 2580$ Ma) link with those of the Shelley Lake granite (2580 Ma, $^{40}Ar/^{39}Ar$, hornblende and biotite) and the Matachewan/Hearst dikes (2450 Ma, U/Pb, zircon and baddeleyite). The sense of motion on the late Archean–early Proterozoic apparent polar wander path for Laurentia, as defined by this sequence of paleopoles, is opposite to that previously accepted.

## 1. Introduction

The Red Lake greenstone belt (RLGB) in the Uchi subprovince (Superior Province, northwestern Ontario) is one of the major gold-producing regions in Canada. Recent structural mapping and U/Pb and $^{40}Ar/^{39}Ar$ dating of the RLGB, its gold deposits and the surrounding batholiths sponsored by the Ontario Geological Survey have improved our understanding of the structural setting, alteration and metamorphism of the Archean supracrustal units and their spatial and temporal relationship to gold mineralization. This work forms the framework for our paleomagnetic study.

Our first purpose was to refine the Archean apparent polar wander path (APWP). The North American APWP has been anchored at its old end by a group of paleopoles from the Abitibi and Wabigoon subprovinces very loosely dated as being $\approx 2700$ Ma old. The units have not been dated radiometrically, nor is there compelling evidence that their natural remanent magnetiza-

[1] Present address: Laboratorio de Geofísica, Departamento de Física, Universidad Simón Bolivar, A.P. 89051, Caracas 1086-A, Venezuela

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tions (NRMs) are primary. The RLGB batholiths mostly have U/Pb and $^{40}Ar/^{39}Ar$ dates of $\geqslant 2700$ Ma [1–6]. Regional cooling following batholith intrusion was relatively fast ($\approx 30°C/Ma$), enhancing the prospects of recovering primary thermoremanent magnetizations (TRMs). If we can indeed recover these TRMs comparison of magnetic blocking temperatures with those determined from $^{40}Ar/^{39}Ar$ would permit precise dating of the paleopoles.

Our second purpose was to study units that were hydrothermally altered during the same event that caused gold mineralization in RLGB supracrustal rocks to see what could be deduced from NRM directions and blocking temperatures about the timing and average temperature of gold formation. NRMs associated with ore deposits are generally chemical remanent magnetizations (CRMs) acquired during hydrothermal alteration and low-grade metamorphism. Such CRMs are usually due to growth and recrystallization of secondary magnetite [7,8] or oxidation (hematization) of magnetite [9–12], although occasionally they are carried by pyrrhotite [11,13,14] or chromite [14]. Our study of altered units in the RLGB is one of the first paleomagnetic investigations of gold mineralization in Archean units of the Canadian Shield. The cooling history of the RLGB derived from U/Pb and $^{40}Ar/^{39}Ar$ separated mineral analyses constrains the age and temperature information derived magnetically.

## 2. Geological setting

The RLGB contains three main supracrustal volcanic units [2]: The lower volcanic sequence of tholeiitic basalts and basaltic komatiites dominates the eastern and western RLGB. The mid-

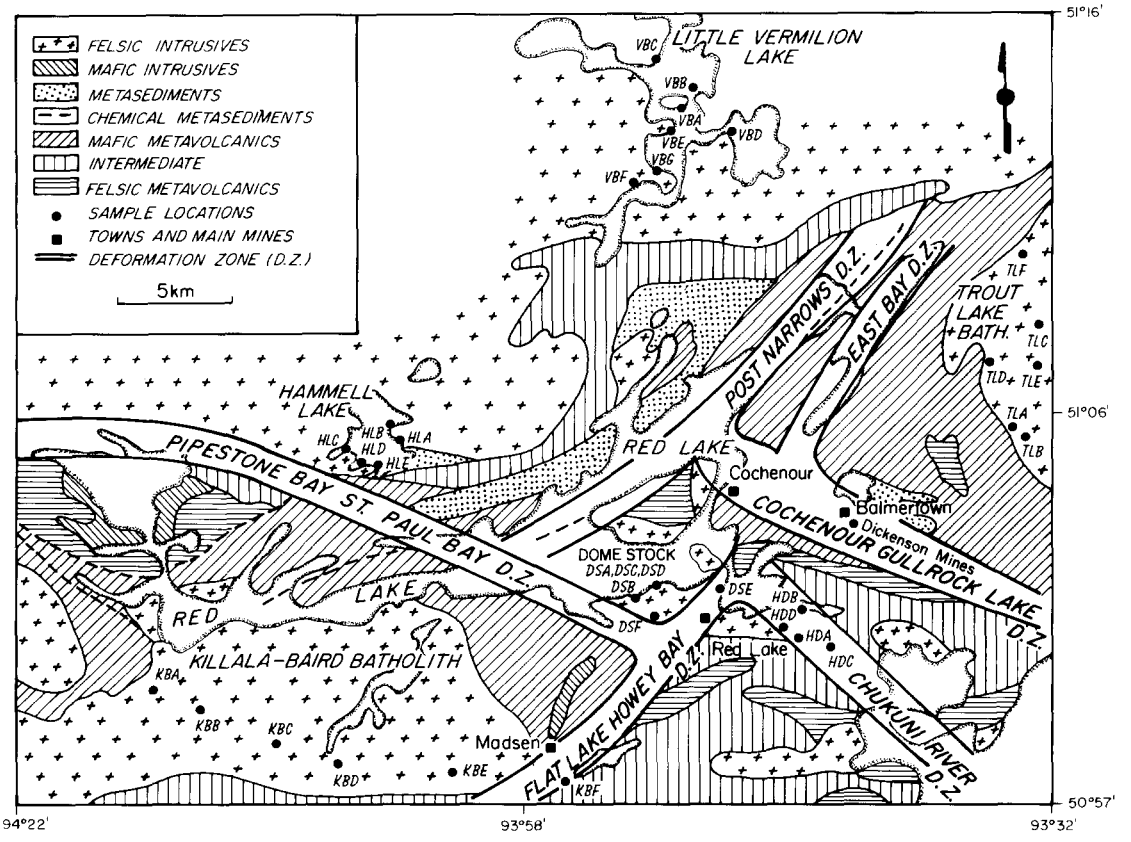



Fig. 1. Major geological units and deformation zones (DZs) in the Red Lake greenstone belt (RLGB), showing sampling sites in the almost unaltered batholiths (VB, HL, TL, KB) and altered units (DS, HD, Dickenson Mines). DZs formed the focus for shearing in the RLGB and also channelled hydrothermal circulation.

dle sequence consists of mafic and felsic pyroclastics. The upper sequence comprises earlier tholeiitic and later calc-alkalic phases. All the supracrustal sequences have been strongly sheared, altered hydrothermally, and metamorphosed to greenschist facies. Their ages span ≈ 260 Ma, from 2992 to 2733 Ma.

Large granitoid batholiths with U/Pb zircon ages between 2730 and 2700 Ma [2] enclose the supracrustal sequences (Fig. 1). By combining isotopic, structural and petrographic results Corfu and Andrews [2] distinguish three phases of granitic plutonism. The first was intrusion of the northern (Little Vermilion and Hammell Lake) batholiths around 2730 Ma. The second phase, coinciding with intrusion of the Trout Lake batholith (2720–2710 Ma), was responsible for much of the alteration, deformation and metamorphism. The final phase (2705–2700 Ma) was associated with the Killala Baird and Howey diorite intrusions in the central part of the RLGB and the marginal phase of the Trout Lake batholith.

Contact metamorphism, alteration, shearing and gold mineralization in the supracrustal units were related sequential events triggered by batholith intrusion. Metamorphic and alteration mineral assemblages suggest that contact thermal metamorphism and hydrothermal alteration occurred simultaneously during a period of active shearing [1]. Two generations of hydrothermal fluids were responsible for alteration: carbonatized metamorphic fluids with background gold values derived from dehydration of supracrustal rocks ahead of the rising granite diapirs, and late magmatic fluids from the batholiths themselves, representing a $H_2O$-dominated regime in which most of the gold precipitation took place [1].

The deformation zones (DZs, 1–3 km wide and 30–50 km long shear and mylonite belts) seem to have been caused by a simple shear regime related to batholith intrusions, extending from early brittle–ductile through ductile and late brittle stages. The deformation zones focused the bulk of hydrothermal fluid flow through the crust, resulting in intense regional-scale alteration and gold deposition. The major gold-producing zones in the RLGB are closely associated with these deformation zones (Fig. 1).

The timing of gold mineralization is still debated. U/Pb zircon and sphene ages of several pre- and post-ore minor intrusive rocks [1,2] suggest an age between 2720 and 2700 Ma. On the other hand, a biotite $^{40}Ar/^{39}Ar$ date from the metamorphic aureole surrounding an ore lens [3] suggests that gold mineralization began around 2640 Ma. Recent $^{40}Ar/^{39}Ar$ data for single grains of hornblende from the Trout Lake batholith indicate a post-orogenic thermal pulse 2655 Ma ago which may be related to gold mineralization [5]. The tail of the cooling curve extends from 2635 to 2570 Ma. A late-stage cooling rate of ≈ 0.3°C/Ma [6] is indicated by different $^{40}Ar/^{39}Ar$ ages for biotite grains of different sizes from the Killala Baird batholith.

## 3. Sampling, measurements and analysis

For paleomagnetic study we sampled the largely unaltered Hammell Lake (HL), Killala Baird (KB), Little Vermilion (VB) and Trout Lake (TL) batholiths and the altered Howey diorite (HD), Dome Stock (DS) and several rock units from the Dickenson Mine (DM) (two quartz-rich felsic dikes, a lamprophyre dike and a quartz–muscovite-rich rhyolitic andesite). HD, DS and DM were hydrothermally altered in the same event that caused gold mineralization.

We collected an average of five samples per site, and oriented them by sun and Brunton compasses. At least one specimen of each sample was demagnetized by an alternating field (AF) in twenty steps to 100 mT and one specimen was thermally demagnetized in ca. fifteen temperature steps, usually to 600°C, using standard Schonstedt demagnetizers. NRM measurements for thermally demagnetized specimens were made with a Schonstedt spinner magnetometer, while a modified Digico spinner magnetometer with similar sensitivity was used for AF-cleaned specimens.

NRM vectors were defined by linear trajectories on orthogonal vector projections [15]. These linear trajectories were tested by principal component analysis [16] and were accepted only if a segment including at least three consecutive points (excluding the origin) had an angular uncertainty of $< 10°$. The characteristic (highest coercivity $H_c$ or laboratory unblocking temperature $T_{UB}$) NRM for each specimen was also obvi-

ous as a stable end-point direction on stereographic projections.

Initial volume susceptibilities $k_o$, which were used to calculate Koenigsberger $Q$ ratios (intensity ratio of NRM/induced magnetization), were measured using a Bartington AC bridge with a peak field of $\approx 0.1$ mT. Saturation isothermal remanent magnetization (SIRM) test samples were initially AF demagnetized to 100 mT, then progressively magnetized in steps to 800 mT using an electromagnet, and finally stepwise AF demagnetized to 100 mT. Their remanences were

TABLE 1

RLN site and grand mean directions

| Site | Decl. (°) | Incl.(°) | $N$ | $\alpha_{95}$ (°) | K |
|---|---|---|---|---|---|
| HDA | 358 | 84 | 6 | 13 | 20 |
| HDB | 351 | 82 | 6 | 20 | 8 |
| HDC | 19 | 84 | 6 | 11 | 27 |
| HDD | 322 | 82 | 6 | 13 | 20 |
| Mean | 350 | 83.4 | 4 | 3 | 714 |
| DSA | 58 | 85 | 6 | 19 | 9 |
| DSB | 70 | 64 | 4 | 24 | 9 |
| DSC | 49 | 46 | 5 | 11 | 33 |
| DSD | 335 | 72 | 5 | 29 | 5 |
| DSE | 241 | 67 | 6 | 21 | 8 |
| DSF | 234 | 81 | 6 | 14 | 17 |
| Mean | 32 | 82 | 6 | 18 | 11 |
| HLA | 288 | 76 | 4 | 17 | 16 |
| HLB | 341 | 70 | 1 | | |
| HLC | 331 | 53 | 1 | | |
| HLD | 57 | 48 | 1 | | |
| HLE | 317 | 67 | 2 | | 93 |
| Mean | 348 | 70 | 5 | 18 | 11 |
| VBA | 357 | 59 | 6 | 18 | 11 |
| VBB | 302 | 76 | 7 | 26 | 7 |
| VBC | 293 | 79 | 5 | 14 | 19 |
| VBD | 0 | 68 | 4 | 16 | 2O |
| VBE | 4 | 76 | 6 | 21 | 7 |
| VBF | 305 | 60 | 5 | 27 | 6 |
| VBG | 73 | 63 | 4 | 30 | 6 |
| Mean | 348 | 74.5 | 7 | 12 | 21 |
| KBA | 323 | 77 | 6 | 14 | 18 |
| KBB | 354 | 79 | 6 | 19 | 9 |
| KBC | 26 | 79 | 4 | 10 | 50 |
| KBD | 312 | 79 | 1 | | |
| KBE | 15 | 48 | 2 | | 64 |
| KBF | 85 | 74 | 5 | 24 | 7 |
| Mean | 10 | 76.4 | 6 | 12 | 23 |
| TLA | 301 | 69 | 5 | 15 | 17 |
| TLB | 350 | 65 | 4 | 30 | 6 |
| TLC | 198 | 64 | 6 | 28 | 4 |
| TLD | 18 | 82 | 5 | 21 | 9 |
| TLE | 135 | 86 | 5 | 13 | 24 |
| TLF | 34 | 80 | 4 | 13 | 29 |
| Mean | 322 | 85 | 6 | 13 | 19 |
| Final mean | 356 | 79 | 6 | 5 | 136 |
| Paleopole | Lat. = 72°N; Long. = 261°E; $dm = 9°$; $dp = 9°$ | | | | |

TABLE 2

RLR site and grand mean directions

| Site | Decl. (°) | Incl.(°) | $N$ | $\alpha_{95}$ (°) | $k$ |
|---|---|---|---|---|---|
| TLA | 143 | −66 | 2 | | 7 |
| TLB | 96 | −45 | 2 | | 210 |
| Mean | 113 | −58 | 2 | | 12.4 |
| VBA | 141 | −63 | 6 | 7 | 60 |
| VBB | 225 | −49 | 4 | 11 | 45 |
| VBD | 195 | −65 | 2 | | 7 |
| VBF | 223 | −53 | 2 | | 19 |
| VBG | 231 | −80 | 2 | | 14 |
| Mean | 205 | −66 | 5 | 15.4 | 16.5 |
| KBC | 142 | −75 | 2 | | 6 |
| KBD | 212 | −60 | 1 | | |
| KBF | 235 | −79 | 4 | 19 | 14 |
| Mean | 199.5 | −74.5 | 3 | 16 | 25 |
| HLA | 67 | −64 | 3 | 29 | 8 |
| HLB | 220 | −65 | 5 | 22 | 8 |
| HLC | 179 | −60 | 3 | 22 | 10 |
| HLE | 76 | −63 | 1 | | |
| Mean | 133 | −77.5 | 4 | 24 | 9 |
| Final mean | 159 | −74 | 4 | 16 | 20 |
| Paleopole | Lat. = 75°N; Long. = 222°E; $dp$ = 26°; $dm$ = 29° | | | | |

measured after each magnetization and demagnetization step.

## 4. Paleomagnetic results and magnetic petrology (Tables 1–3)

### *4.1 RLN and RLR magnetizations*

The Red Lake samples often exhibit a low-$H_c$, low-$T_{UB}$ NRM with normal (N) polarity (RLN) whose average direction is $D = 356°$, $I = 79°$. RLN decays exponentially during AF cleaning and has few coercivities > 20 mT. It is often a low-$T_{UB}$ overprint superimposed on other remanences. In most cases, RLN is probably a Brunhes-epoch viscous remanent magnetization (VRM).

The southerly, steeply up RLR magnetization is the characteristic (highest $H_c$ and $T_{UB}$) NRM of the batholiths. Its average direction is $D =$

TABLE 3

RLG site and grand mean directions

| Site | Decl. (°) | Incl.(°) | $N$ | $\alpha_{95}$ (°) | $k$ |
|---|---|---|---|---|---|
| HDA | 204 | −61 | 4 | 15 | 21 |
| HDC | 169 | −54 | 1 | | |
| Mean | 185 | −59 | 2 | | 34 |
| DSA | 184 | −40 | 3 | 18 | 20 |
| DSC | 202 | −51 | 6 | 9 | 4 |
| DSE | 124 | −70 | 2 | | 8 |
| Mean | 180 | −57 | 3 | 24 | 11 |
| DM | 176 | −56 | 4 | 34 | 4 |
| Final mean | 180 | −557 | 3 | 3 | 23 |
| Paleopole | Lat. = 77°N; Long. = 86°E; $dp$ = 3°; $dm$ = 4° | | | | |

159°, $I = -74°$, almost 180° reversed to RLN. In thin section (Fig. 2a and b) the silicate minerals in these rocks show no signs of alteration or metamorphism and the visible opaque minerals are euhedral and pristine. Magnetite and hematite, the latter sometimes with lens-like inclusions of ilmenite lamellae, are the main opaques. Magnetite is an abundant primary mineral in granitic intrusions associated with gold deposits in the Superior Province [17]. Some of the hematite may have crystallized directly from the magma [18], but most is probably a high-temperature deuteric oxidation product. The cooling rate of the batholiths ($\approx 30°C/Ma$ [3]) was slow enough to allow deuteric oxidation to reach completion in the 1000–600°C range [19]. Of course, there is no guarantee that the fine, optically invisible opaques which carry much of the RLR remanence have not been altered. However, as we discuss in the next paragraph, the same RLR remanence is carried in some cases by coarse-grained magnetite.

Figure 3 shows the AF and thermal demagnetization response of the RLR remanence in a VB sample. A soft RLN remanence with coercivities of mainly $< 20$ mT and $T_{UB} < 350°C$ is stripped away to reveal the underlying hard RLR remanence, more than half of which is carried by hematite with $H_c > 100$ mT and $T_{UB} > 580°C$; the remainder is carried by fine-grained magnetite with $300°C \leqslant T_{UB} \leqslant 580°C$. In an HL sample (Fig. 4), RLR is the only significant magnetization.

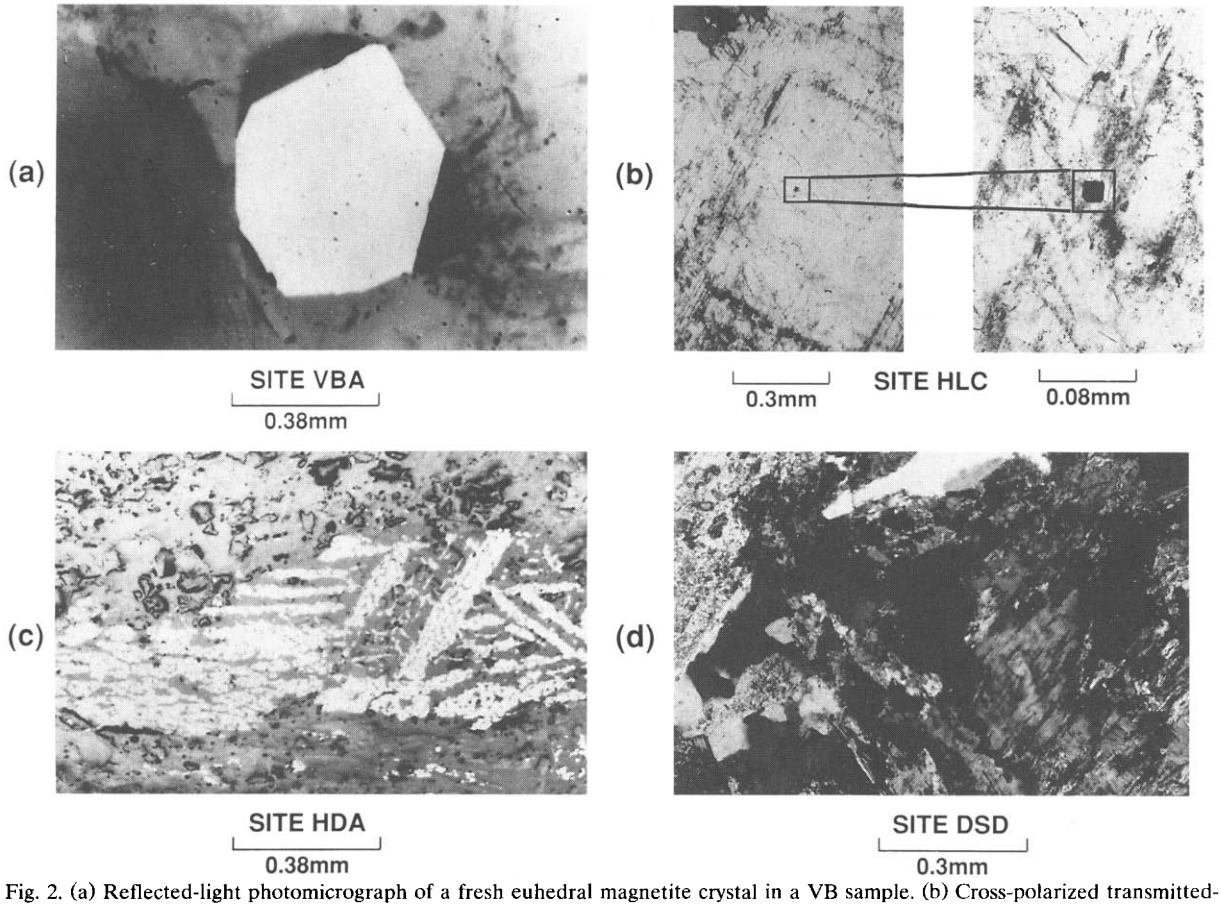



Fig. 2. (a) Reflected-light photomicrograph of a fresh euhedral magnetite crystal in a VB sample. (b) Cross-polarized transmitted-light photomicrograph of a plagioclase crystal from a HL sample, which has trapped a euhedral opaque crystal (probably magnetite) during its growth. Ghost lines of impurities record different stages of growth. Reflected-light (c) and (d) transmitted-light photomicrographs of altered opaques and silicates in HD and DS samples (see discussion in text). The fresh unaltered opaques and silicates in (a) and (b) contrast with the very altered minerals in (c) and (d).

The remanence decays exponentially with AF cleaning and unblocking temperatures are uniformly distributed from 150 to 580°C. In order to have these magnetic properties, the magnetites must be of multidomain size (> 10–15 $\mu$m) and therefore optically visible.

### *4.2 RLG magnetization*

Sheared and altered units (DS, HD, DM) from the gold-bearing DZs preserve a reverse-polarity remanence (RLG) with the mean direction $D = 180°$, $I = -57°$. Samples from these units are heavily altered in thin section (Fig. 2c and d). Primary mineralogies are almost completely obliterated. For example, a HD sample (Fig. 2c) has relict titanomagnetite oxyexsolution textures, but the residual magnetite has altered to hematite, while the ilmenite has been replaced by rutile. Browning and Beske-Diehl [12] report similar hematized ilmenite oxyexsolution lamellae in altered flow interiors of the middle Proterozoic Portage Lake volcanics (Michigan). In their samples, a secondary CRM carried by hematite constrains the age of copper mineralization in the region. A DS sample (Fig. 2d) has been pervasively hydrothermally altered. Anhedral opaque crystals appear in a background of dusty altered feldspars and chloritized micas.

Figure 5 gives examples of AF and thermal cleaning of the RLG remanence in samples from the Dome Stock. In DSC-6, the dominant RLG remanence, with coercivities characteristic of fine-grained, single-domain magnetite, is overprinted by a minor soft N-polarity NRM. DSA-4 has a larger soft overprint, but the RLG rema-

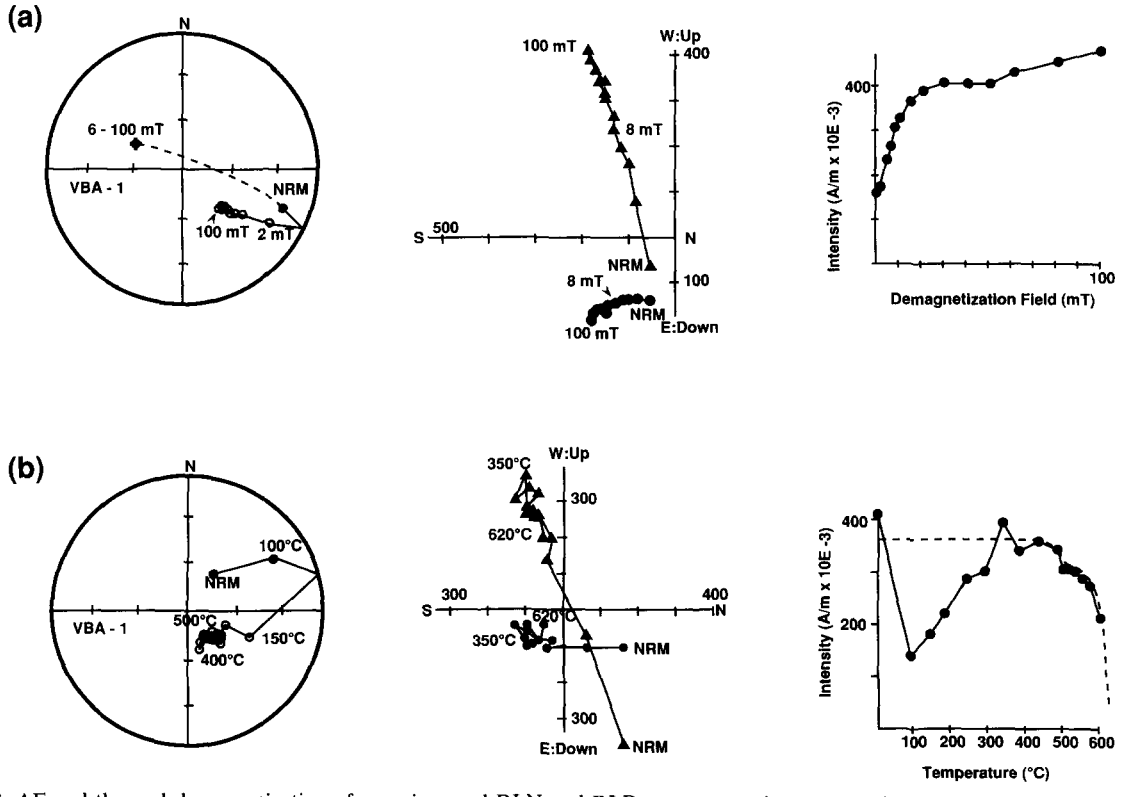



Fig. 3. AF and thermal demagnetization of superimposed RLN and RLR remanences in a sample from the VB batholith. In the equal-area projections, open and closed symbols are upper and lower hemisphere vectors respectively, and the cross is the vector subtracted between 6 and 100 mT, which estimates the RLN direction. In the orthogonal vector projections, • and △ denote horizontal and vertical plane projections respectively. In the intensity plots, solid curves represent the total NRM vector (RLN + RLR) and dashed curves (calculated by the method of Dunlop [22]) represent separated vectors, in this case RLR.

nence is isolated above 350°C. About one-third of RLG resides in hematite and the remainder in fine-grained magnetite with distributed $T_{UB}$s.

## 5. Magnetic properties: SIRM, NRM, susceptibility and Q ratio

SIRM acquisition and AF demagnetization curves for representative samples carrying RLN, RLR or RLG magnetizations are shown in Fig. 6. In RLN samples (Fig. 6a) the IRM saturates and subsequently demagnetizes in rather low fields. The SIRM carrier is probably coarse multidomain magnetite. In some RLG samples (Fig. 6b) IRM is acquired gradually, remains far from saturation at 100 mT, and is very resistant to AF cleaning. Such behavior is characteristic of hematite or possibly fine, single-domain magnetite. In RLR samples (Fig. 6c and d), the SIRM has mainly soft carriers (multidomain magnetite), with isolated examples of very hard carriers (probably hematite).

Intensities of NRM ($M_{NRM}$) and induced magnetization ($k_0 H$, for $H = 0.06$ mT) measured for 48 specimens are plotted in Fig. 7. Contours of the Koenigsberger $Q$ ratio ($Q_n = M_{NRM}/k_0 H$) are lines of unit slope. Specimens with univectorial RLR or RLG remanences (Fig. 7a) have a large fraction of NRM carriers with $Q_n > 1$, while those with RLN overprints have almost exclusively carriers with $Q_n < 1$. This is not surprising, since we interpret RLN to be a VRM in most cases and most of the magnetic mineral content of felsic plutonic rocks is in coarse, multidomain opaques, which tend to be soft and viscous [20,21]. The potassium and sodium feldspars of granitic rocks lack the magnetic inclusions responsible for the intense (and hard) NRMs and high $Q$ ratios of many mafic and intermediate plutonic rocks.

Nevertheless, many of our samples do retain high-coercivity NRMs, which must reside in a small fraction of single-domain magnetite and hematite that is undetected by bulk magnetic properties such as susceptibility and SIRM. Figure 8 illustrates that RLN, RLR and RLG remanences are distinctively different in both NRM intensity and median AF destructive field. (In the case of composite or multivectorial NRMs, the

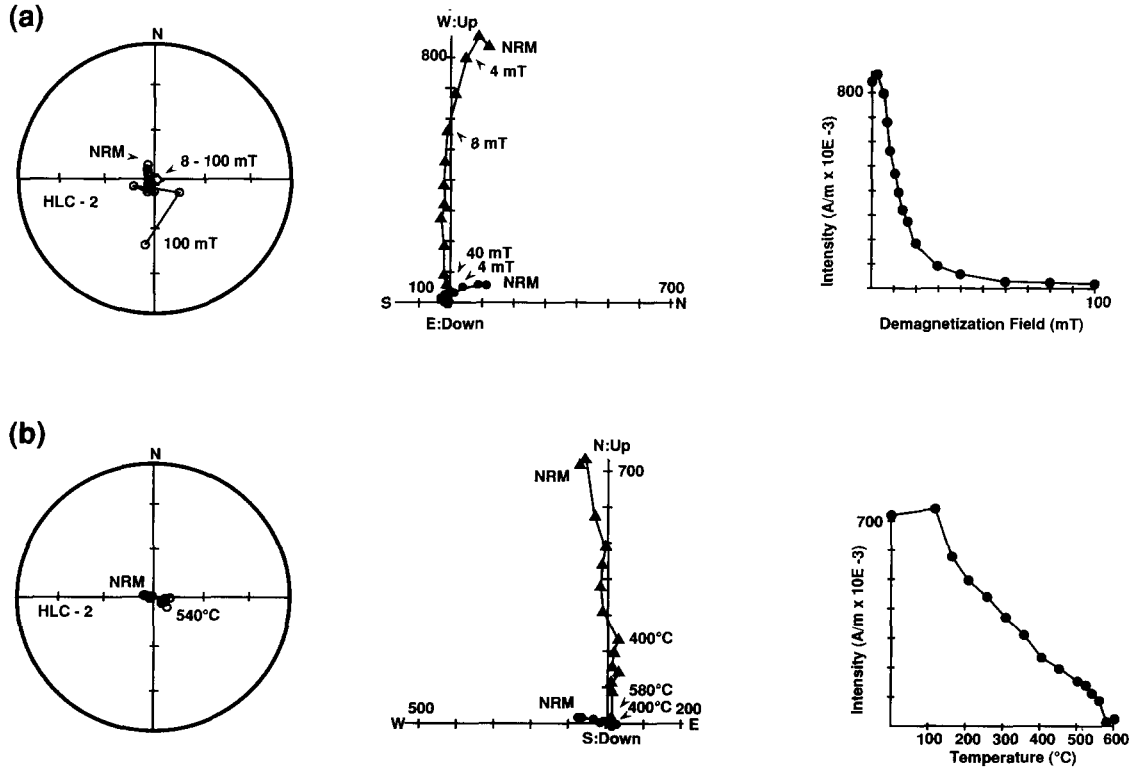



Fig. 4. AF and thermal demagnetization of the RLR magnetization in a sample from the HL batholith. Symbols as in Fig. 3.

intensities plotted are those of the separated components.) RLR and RLG are much harder than RLN. On average, RLG has slightly higher median destructive fields than RLR, but the ranges overlap. However, the difference in NRM intensities is striking. RLG averages almost two orders of magnitude weaker than RLR. We interpret these results to mean that both RLR and RLG are stable (high $H_c$, high $T_{UB}$) ancient remanences, but that alteration has destroyed all but a small fraction of the original stable remanence carriers in the RLG samples.

## 6. Discussion

### 6.1 *Interpretation of RLN, RLR and RLG magnetizations*

Figure 9 shows site, unit and grand means for the RLN, RLR and RLG magnetizations. RLN directions scatter around the direction of the present earth field and are probably mostly Brunhes-epoch VRMs, as discussed earlier. Some RLN magnetizations have implausibly high coercivities for magnetite VRMs, although they could be hematite VRMs. Alternatively they could be normal-polarity counterparts of RLR. Seeing as there are few of them and their origin is ambiguous, we will not consider them further.

The rocks carrying RLR magnetizations show no signs of later alteration, and the RLR direction does not resemble younger Precambrian or normal-polarity Phanerozoic field directions. In principle, the RLR could be a low-temperature CRM of late Tertiary age, but the high coercivities and exclusively reverse polarities make this unlikely. RLG samples should be equally susceptible to a Tertiary remagnetization, but very few of them have RLR components. Instead we argue that RLR is a primary TRM dating from initial cooling of the batholiths. Some of the within-unit and between-unit dispersion of RLR directions

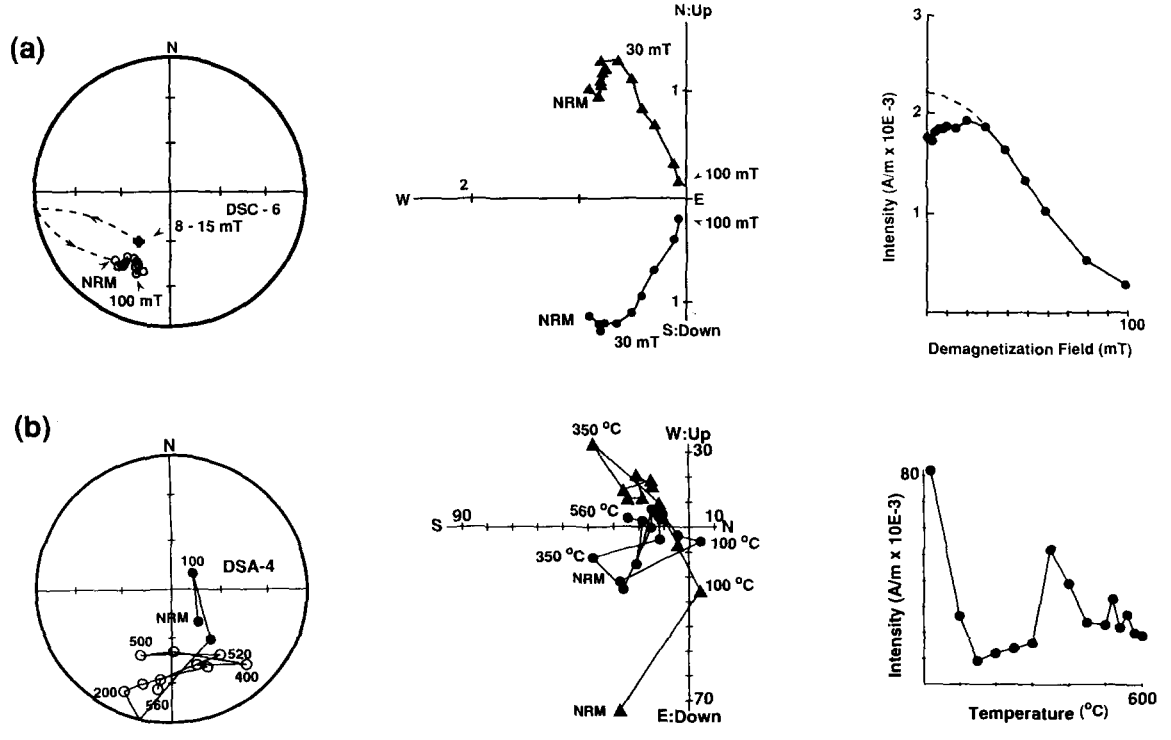



Fig. 5. AF and thermal demagnetization of superimposed RLN and RLG remanences in altered samples of the Dome Stock. Numbers on the equal-area projection and vector plot in (b) are temperatures (°C). In the intensity plot in (a), the decay curve of the separated RLG remanence is shown dashed. The intensity decay in (b) is too noisy to permit a good separation of vectors, but the vector plot shows that RLN is completely removed by 350°C, while about two-thirds of the RLG vector demagnetizes between 350 and 580°C. Symbols as in Fig. 3.

may be due to the shortcomings of granitic rocks as magnetic recorders, but much of it probably results from apparent polar wander during slow cooling and from differences in intrusion age (and hence mean magnetization age) from one batholith to another, which could be as much as 30 Ma (see section 2). If we were to exclude the Trout Lake batholith, whose different intrusive phases differ in age by at least 15 Ma [2], the precision of the grand mean would increase markedly. However, because of our present uncertainty about exact magnetization ages and cooling histories we prefer to include RLR results from all the batholiths in a single average.

The RLG mean direction is similar in declination to RLR but much shallower in inclination (Fig. 9b). Rocks carrying the RLG magnetization are severely altered and their mineralogies, including the opaque minerals, have been almost totally reconstituted. The RLG remanence has high average coercivities but is on average between 1 and 2 orders of magnitude less intense than the RLR remanence (Fig. 8). We believe that RLG is a secondary CRM or thermochemical remanent magnetization (TCRM). The timing of alteration and of acquisition of RLG is still debated, U/Pb suggesting $\approx 2700$ Ma [2] and $^{40}Ar/^{39}Ar$ favoring $\approx 2650$ Ma [5] or even as late as 2570 Ma [6], corresponding to the low-temperature tail of the cooling curve.

Has any primary RLR remanence survived in altered samples? In view of the severity of the alteration, it is not surprising that only a few specimens show coexisting RLR and RLG magnetizations. One example is sample DSC-5 from the Dome Stock (Fig. 10), in which a steeply inclined RLR vector is demagnetized over the 2–60 mT range and a shallower RLG remanence is erased between 60 and 100 mT. Both magnetizations seem to reside in single-domain or pseudo-single-domain magnetite with comparatively high coercivities (in Fig. 10, intensity decay

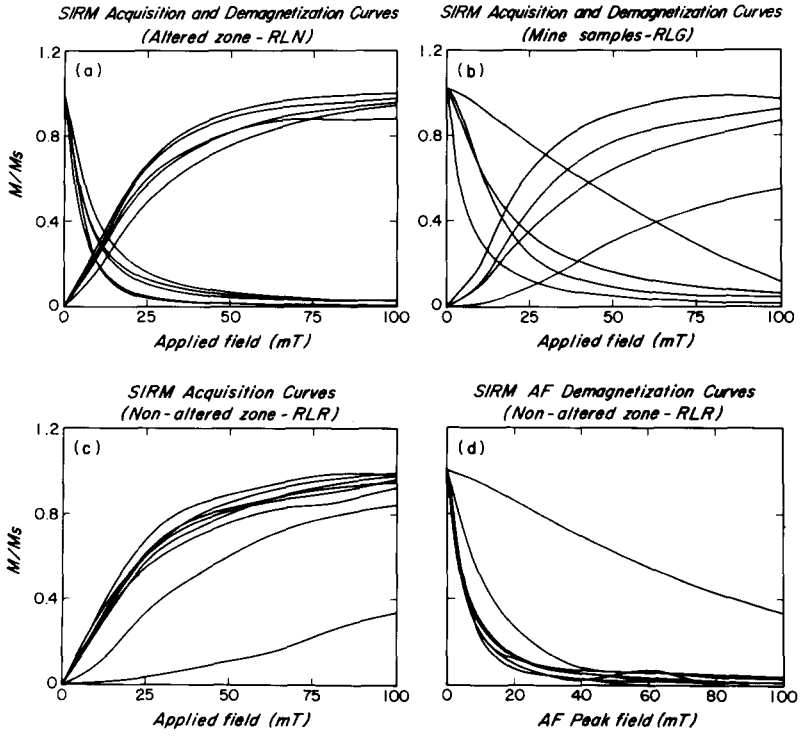



Fig. 6. SIRM acquisition and AF demagnetization curves for samples from the altered and unaltered zones in the Red Lake area. Coercivities are much higher for most RLG samples than for those carrying RLN or most RLR magnetizations.

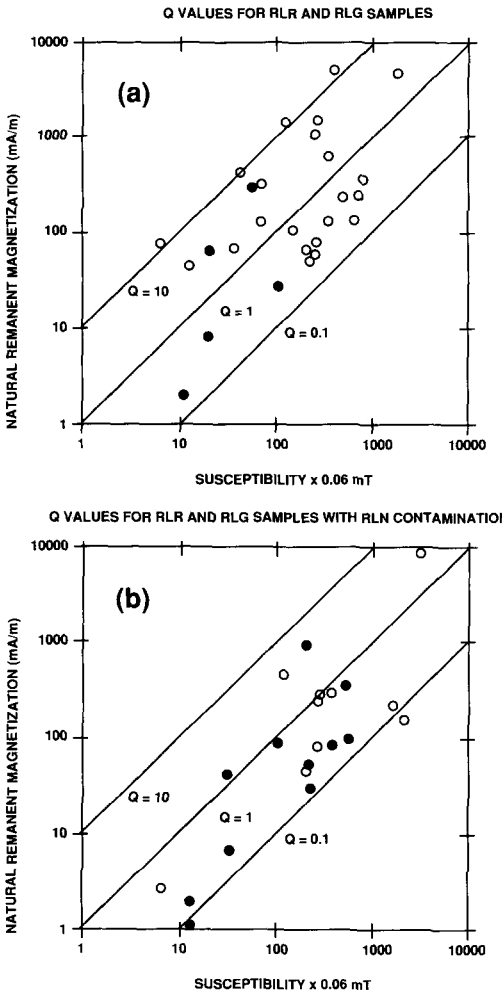



Fig. 7. NRM vs. induced magnetization diagrams for (a) samples carrying univectorial RLR (○) or RLG (●) and (b) samples with RLN overprinting either RLR or RLG. Altered samples carrying RLG have lower NRM intensities and susceptibilities than RLR samples. The diagonal lines are contours of constant Koenigsberger $Q_n$ ratio.

spectra for the components separately were calculated from the Zijderveld vector plot, as described by Dunlop [22]).

### 6.2 *Paleomagnetic poles for RLR and RLG*

The paleomagnetic pole corresponding to the RLR grand mean falls at 75°N, 222°E (Fig. 11), among ≈ 2740–2680 Ma paleopoles for the Otto Stock [23–25] and several plutons in the Meens Lake–Dempster Lake greenstone belt [26,27]. This age range is appropriate, since RLR is the

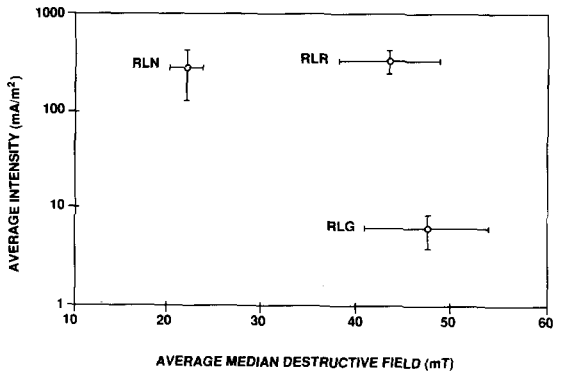



Fig. 8. Plot of NRM intensities and median destructive fields for separated RLN, RLR and RLG magnetizations. Error bars represent $\pm 1\sigma$. RLN is an intense but soft remanence, probably in most cases a modern VRM. RLR and RLG are both hard remanences, but RLG is weaker by a factor 50 than RLR. We interpret RLR to be a TRM carried by unaltered primary magnetite and hematite and RLG to be a secondary CRM acquired after destruction of most of the primary minerals.

primary TRM of batholiths with U/Pb ages of 2730–2700 Ma [2]. A more precise age is possible by comparing typical RLR blocking temperatures (corrected for slow cooling in nature [e.g., 28]) to the preliminary cooling curves of McMaster [3] and Wright et al. [6] which are based on U/Pb and $^{40}Ar/^{39}Ar$ data.

The paleopole corresponding to the RLG grand mean falls at 77°N, 86°E. The RLG magnetization is characteristic of the Dome Stock, Howey diorite and Dickenson mine samples from the central part of the RLGB, which is transected by deformation zones (DZs) where shear stress and deformation and hydrothermal alteration were concentrated. The DZs also host the gold-

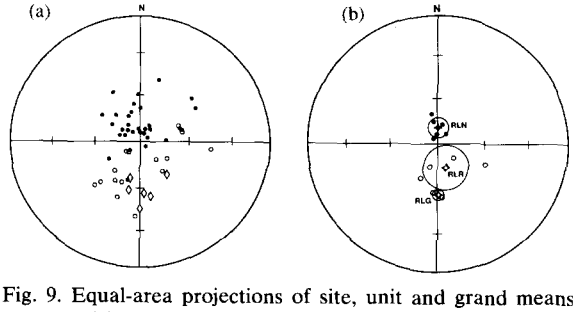



Fig. 9. Equal-area projections of site, unit and grand means for RLN (●), RLR (○) and RLG (◇in (a), ○in (b)). Circles of 95% confidence about the grand means are given in (b).

producing deposits. The Flat Lake–Howey Bay DZ (Fig. 1) deformed the 2718 Ma Dome Stock but became inactive before intrusion of dikes cutting the Howey diorite 2699 Ma ago. Shearing on this DZ was contemporaneous with thermal metamorphism related to the nearby Killala Baird batholith [1,2], but the isotopic cooling curve shows that the thermal effects endured long after 2700 Ma.

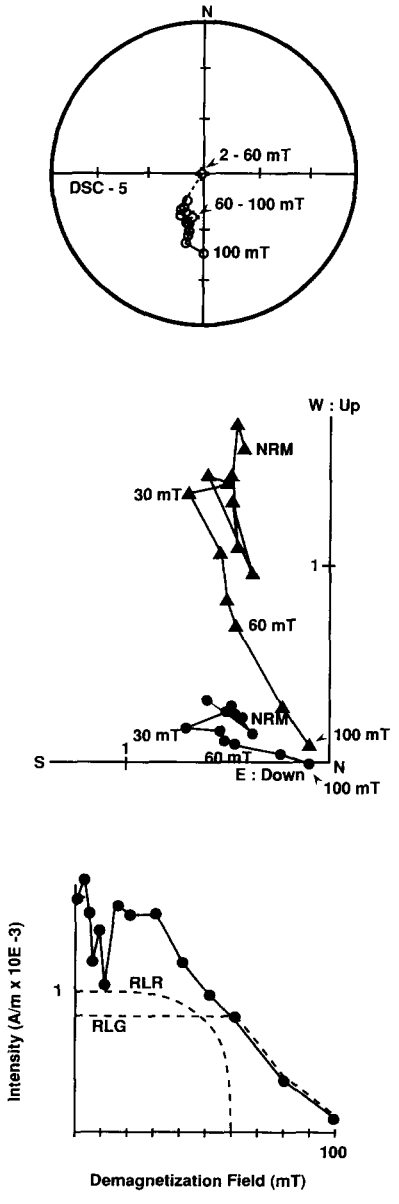



Fig. 10. AF demagnetization of an altered sample from the Dome Stock in which RLR and RLG are superimposed. Symbols as in Fig. 3. The separated RLG and RLR decay curves (dashed) show that RLR and RLG have coercivities that are, respectively, below and above 60 mT in this sample.

We propose that the mineralogy of the DS, HD and DM samples was reconstituted at or just before 2700 Ma during active shearing and hydrothermal alteration, but that the RLG magnetization was acquired later during regional cooling as a relatively low-temperature CRM or TCRM. RLG might, for example, correspond to the 2635–2570 Ma (290–255°C) tail of the RLGB cooling curve [6]. The difference of 100–150 Ma in acquisition times of RLR and RLG magnetizations accounts for the difference between the RLR and RLG paleopoles. Gold mineralization was also a late, relatively low-temperature event [3,5], and would correspond more nearly in age to RLG than RLR.

The RLG paleopole is indistinguishable from pole SL1 for the Shelley Lake granite primary remanence [29], whose age is $2580 \pm 20$ Ma by cross-calibration to $^{40}Ar/^{39}Ar$ hornblende and biotite ages [30]. Therefore, it is likely that RLG has an age of about 2580 Ma and that the RLG magnetization was acquired at a temperature of approximately 250°C, on the tail of the regional cooling curve [6]. This temperature estimate for RLG falls in the range 350–200°C that is typical of gold mineralization, as determined from fluid inclusion studies and from K-Ar analyses of hydrothermal illites associated with gold formation [31,32]. A temperature of 250°C is also consistent with the observed low-grade metamorphism.

### *6.3 The late Archean–early Proterozoic APWP of Laurentia*

The RLR and RLG paleopoles, if our proposed ages of ≈ 2700 Ma and ≈ 2580 Ma respectively are correct, suggest that the previously accepted late Archean APWP must be revised. The recent demonstration that the Matachewan–Hearst dike swarm is early Proterozoic (2450 Ma, U/Pb on baddeleyite and zircon [33]; 2480 Ma, U/Pb baddeleyite isochron [34]) rather than Archean (2690 Ma, whole-rock Rb/Sr [35]) as previously believed changes the time calibration of the pivotal MD paleopole [36–40] and confirms that the APWP (Fig. 11) must young from west to east rather than from east to west as in

most previously published APWPs [e.g., 29,36,41–43].

Buchan et al. [24] have partly addressed this problem by connecting pole OS for the 2680 Ma (U/Pb zircon [25]) Otto Stock to the ≈ 2450 Ma MD poles. The OS magnetization has not been shown to be primary, but Buchan et al. have proven that it is older than 2450 Ma. Symons and Vandall [44] and Vandall and Symons [45] have incorporated both dated tiepoint paleopoles SL1 and MD in the APWP by following the conventional track from assumed primary Archean Abitibi and Wabigoon subprovince poles (KK, GR, AB, DS and WB, which plot in or near Africa in Fig. 11) across Europe to SL1 at 2580 Ma and then reversing the motion and 'backtracking' to MD at 2450 Ma. It is not unusual for continental motions to reverse, causing backtracking on the APWP: 'cusps' in the Phanerozoic path for North America [e.g., 46] and 'hairpins' in the Proterozoic path [47,48] are well documented. But the problem of two different groups of paleopoles claimed to be ≈ 2700 Ma old has not been addressed by any of the previous authors.

We will consider first the group of poles labelled 2700 Ma? in Fig. 11. Two of these (KK, Kamiskotia gabbro complex [36], and AB, Abitibi Belt gabbros and basalts [37]) are from Abitibi subprovince units with formation ages in the range

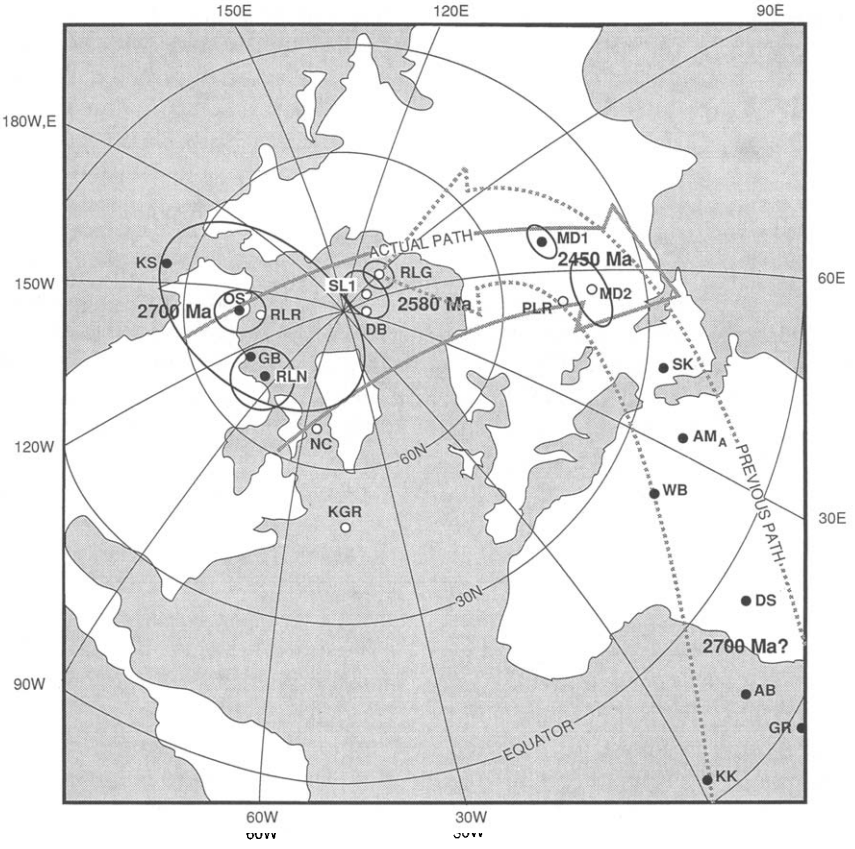



Fig. 11. Two versions of the late Archean and early Proterozoic APWP for North America. The redefined APWP links paleopoles RLR (≈ 2700 Ma, this study), OS (Otto Stock, 2680 Ma, pole not shown to be primary [23–25]), SL1 (Shelley Lake granite, 2580 Ma [29,30]), RLG (≈ 2580 Ma, this study) and MD1 and MD2 (Matachewan dikes, 2454 Ma [33]). Also shown are the RLN pole (present study) and ≈ 2700 Ma old poles DB, KGR, KS, GB and NC (Dobie Lake batholith, Kawashe Lake gabbro, Kawashe Lake Stock, Graniteboss Lake Stock and North Caribu Lake batholith respectively [26]). ○ and ● = reversed and normal polarity. Ovals of 95% confidence are shown for the paleopoles determined in this study (RLR, RLG and RLN) and for the well-dated tiepoint poles OS, SL1 and MD1 and MD2. The previously accepted late Archean–early Proterozoic APWP is anchored by the loose swath of poles marked 2700 Ma? (KK, GR, AB, DS, WB, AM$_A$, SK [36,37,49,51,54,55]) and trends in the opposite sense.

2727–2689 Ma (U/Pb zircon [25]). Positive contact tests were reported for Matachewan dikes cutting the gabbros in both studies, implying that the gabbro magnetizations are older than 2450 Ma, although not necessarily primary. However, there is room for doubt. The KK contact test was incomplete in that samples were taken at the dike contact and at large distances from the contact, but not in the intermediate 'hybrid zone' of partial thermal overprinting (see discussion in [24]). Hybrid zone samples with bivectorial magnetizations were studied in the AB gabbro, but there was a progressive change in magnetic properties from the baked rocks near the contact to the unbaked gabbros at large distances. Thus, instead of simple thermal overprinting, in which one remanence is partially replaced by another, we may be dealing with two juxtaposed magnetizations carried by two different generations of magnetic minerals.

In the same group is pole WB for the Wabigoon gabbro [49], which was argued to be primary on the basis of reversals believed to mark the primary cooling front in the intrusion; its age would be in the 2740–2700 Ma range by association with ages of related units in the Wabigoon subprovince [50]. KK, AB and WB, as well as other poles in the first group, could be accommodated on the middle Proterozoic APWP by reversing their polarity. There is in fact evidence for magnetic overprints of pre-Keweenawan and Keweenawan age (1250–1100 Ma) in some units in or adjacent to the Abitibi [51–53] and in the Quetico and Wawa–Shebandowan subprovinces immediately south of the Wabigoon [29,42] that resemble in direction the remanences from which these group 1 paleopoles are derived. A Keweenawan-age remanence also overprints Matachewan dikes in the Wawa subprovince in north-central Ontario [H.C. Halls, pers. commun., 1992]. A similar direction of remanence reported by Vandall and Symons [45] from the Baldhead River quartz monzonite in the Michipicoten belt of the Wawa could possibly be a Keweenawan overprint, although Vandall and Symons interpret it as a late Archean remanence.

The second group of poles, marked 2700 Ma in Fig. 11, includes RLR and OS, as well as a rather scattered set of poles (DB, KGR, KS, GB, NC [26]) from mafic and felsic plutons in the Meens Lake–Dempster Lake greenstone belt, northwestern Superior Province, with intrusion ages from 2747 to 2720 Ma [27]. The ages of the Red Lake batholiths are in a similar range (2730–2700 Ma) and the Otto Stock is slightly younger (2680 Ma). There is no direct proof that these poles are primary, except for the observation that RLG is consistent with the 2580 Ma pole SL1, and the altered rocks which carry RLG show no sign of overprinting by the RLR remanence that is so conspicuous in the unaltered batholiths. Unlike group 1 poles, the group 2 poles cannot be made to fall on any younger part of the APWP. Furthermore, although they fall close to the present north pole, they cannot be dismissed as Brunhes overprints because most of them, namely RLR, DB, KGR, KS and GB, are based on reversed, or mixtures of reversed and normal, magnetizations.

The question of which group of $\approx 2700$ Ma poles is actually primary is not conclusively settled, but there are increasing numbers of poles from relatively undisturbed Archean units with no sign of remagnetization that fall in the RLR, OS, etc. grouping. We will tentatively take this to be the true 2700 Ma pole position for Laurentia. The segment of the late Archean–early Proterozoic APWP between 2580 and 2450 Ma is now reliably age calibrated. RLG falls on the older end of this segment, and if RLR is in fact older than RLG as we believe, the APWP can be extended back in time another $\approx 100$ Ma.

## 7. Conclusions

(1) RLR, the characteristic magnetization of the unaltered Red Lake batholiths, is probably a primary TRM acquired $\approx 2700$ Ma ago ($^{40}Ar/^{39}Ar$ initial cooling age [3]) and carried by both magnetite and deuteric hematite. Evidence for a primary TRM origin includes its generally high unblocking temperatures ($> 550°C$), coercivities ($> 40$ mT), and NRM intensities ($> 100$ mA/m), as well as the pristine condition of the visible silicate and opaque minerals in thin section.

(2) RLG, the characteristic magnetization of sheared, altered and metamorphosed units in and near the deformation zones (DZs), is a secondary CRM or TCRM $\approx 2580$ Ma old. The age is based both on the similarity of the RLG paleopole to dated pole SL1 (Fig. 11) and cross-calibration of

RLG to the $^{40}Ar/^{39}Ar$ cooling curve [6], assuming the temperature range for acquisition of RLG matches that of gold mineralization in related supracrustal units. If both lines of reasoning are sound, RLG was acquired at a temperature of $\approx 250°C$ in the final stages of regional cooling. RLG is carried mainly by secondary hematite and fine-grained magnetite. The high coercivities of SIRM compared to RLR samples as well as the generally low intensities of NRM ($< 10$ mA/m) and susceptibility imply that primary magnetic minerals have been largely destroyed and replaced by secondary oxides, as is obvious also from thin sections.

(3) RLG provides one of the first paleomagnetically dated records of a Precambrian mineralization event. Since RLG characterizes the shear zone rocks but is absent from unaltered rocks outside the DZs, it could be used as a rapid prospecting tool in delineating areas of alteration and possible mineralization.

(4) Pole RLR is one of the oldest North American paleopoles, and one of the few Archean poles that can be $^{40}Ar/^{39}Ar$ dated.

(5) The RLR and RLG paleopoles define a 2700–2580 Ma segment of the APWP for Laurentia that tracks from North America towards central Asia through the north pole (Fig. 11). The young end of this path segment is also tied by the 2580 Ma paleopole SL1 for the Shelley Lake granite ($^{40}Ar/^{39}Ar$ hornblende and biotite age by Berger and York [30]), and the APWP can be continued in a simple fashion towards India to include the 2450 Ma MD paleopoles for the Matachewan/Hearst dike swarm (U/Pb zircon age by Heaman [33]). The direction of younging along this path is opposite to that previously accepted for late Archean and early Proterozoic time.

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## References

1 A.J. Andrews, H. Hugon, M. Durocher, F. Corfu and M.J. Lavigne, The anatomy of a gold bearing greenstone belt: Red Lake Northwestern Ontario, Canada, in: Proc. Gold'86 Symp., A.J. McDonald, ed., pp. 3–22, 1986.
2 F. Corfu and J.A. Andrews, Geochronological constraints on the timing of magmatism, deformation and gold mineralization in the Red Lake greenstone belt, northwestern Ontario, Can. J. Earth Sci. 24, 1302–1320, 1987.
3 D. McMaster, A preliminary $^{40}Ar/^{39}Ar$ study of the thermal history and age of gold in the Red Lake greenstone belt, M.Sc. Thesis, Univ. Toronto, 1987.
4 S.R. Noble, Geology, geochemistry and isotope geology of the Trout Lake batholith and the Uchi–Confederation Lakes greenstone belt, northwestern Ontario, Ph.D. Thesis, Univ. Toronto, 1989.
5 N. Wright and D. York, Thermal history of the Trout Lake batholith (Red Lake, Ontario) via $^{40}Ar/^{39}Ar$ single-crystal analysis of biotites and hornblendes (abstract), Eos 71, 1297, 1990.
6 N. Wright, P.W. Layer and D. York, New insights into thermal history from single grain $^{40}Ar/^{39}Ar$ analysis of biotite, Earth Planet. Sci. Lett. 104, 70–79, 1991.
7 V. Bachtadse, R. Van der Voo, F.M. Haynes and S.E. Kesler, Late Paleozoic magnetization of mineralized and unmineralized Ordovician carbonates from east Tennessee: evidence for a post-ore chemical event, J. Geophys. Res. 92, 14165–14175, 1987.
8 H. Pan, D.T.A. Symons and D.F. Sangster, Paleomagnetism of the Mississippi Valley-type ores and host rocks in the northern Arkansas and Tri-state districts, Can. J. Earth Sci. 27, 923–931, 1990.
9 H. Ueno, Duration of the Kuroko mineralization episode, Nature 253, 428–429, 1975.
10 J.W. Geissman, W.C. Kelly, R. Van der Voo and G.H. Brimhall, Jr., Paleomagnetic documentation of the early, high-temperature zone of mineralization at Butte, Montana, Econ. Geol. 75, 1210–1219, 1980.
11 H. Ueno and S. Tonouchi, Paleomagnetic evidence for the timing of formation of the Chichibu pyrometasomatic deposits, Japan, Econ. Geol. 82, 1723–1731, 1987.
12 T.D. Browning and S.J. Beske-Diehl, Paleomagnetism and the age of copper mineralization in the Portage Lake Volcanics, Upper Peninsula, Michigan, Can. J. Earth Sci. 24, 2396–2404, 1987.
13 J. Urrutia-Fucugauchi and Z. Jurado-Chichay, Paleomagnetic correlation and dating of mineralization and intrusive events, Phys. Earth Planet. Inter. 55, 65–78, 1989.
14 E. Refai, N.A. Wassif and A. Shoaib, Stability of rema-

nence and paleomagnetic studies of some chromite ores from Barramiya and Allawi occurrences, eastern desert, Egypt, Earth Planet. Sci. Lett. 94, 151–159, 1989.
15 J.D.A. Zijderveld, A.C. demagnetization in rocks: analysis of results, in: Methods in Paleomagnetism, D.W. Collinson, K.M. Creer and S.K. Runcorn, eds., pp. 254–286, Elsevier, New York, 1967.
16 J.L. Kirschvink, The least-squares line and plane and the analysis of paleomagnetic data, Geophys. J.R. Astron. Soc. 62, 699–718, 1980.
17 K. Hattori, Magnetic felsic intrusions associated with Canadian Archean gold deposits, Geology 15, 1107–1111, 1987.
18 W.A. Deer, R.A. Howie and J. Zussman, An Introduction to the Rock-forming Minerals, Wiley, 1982.
19 R.S. Carmichael, Magnetic properties of minerals and rocks, in: Handbook of Physical Properties of Rocks, R.S. Carmichael, ed., Vol. 2, pp. 229–288, CRC Press, 1982.
20 D.J. Dunlop, Determination of domain structure in igneous rocks by alternating field and other methods, Earth Planet. Sci. Lett. 63, 353–367, 1983.
21 D.J. Dunlop, L.D. Schutts and C.J. Hale, Paleomagnetism of Archean rocks from northwestern Ontario III: Rock magnetism of the Shelley Lake granite, Quetico subprovince, Can. J. Earth Sci. 21, 879–886, 1984.
22 D.J. Dunlop, On the use of Zijderveld vector diagrams in multicomponent paleomagnetic studies, Phys. Earth Planet. Inter. 20, 12–24, 1979.
23 G. Pullaiah and E. Irving, Paleomagnetism of the contact aureole and late dikes of the Otto Stock, Ontario, and its application to early Proterozoic apparent polar wandering, Can. J. Earth Sci. 12, 1609–1618, 1975.
24 K.L. Buchan, D.J. Neilson and C.J. Hale, Relative age of Otto Stock and Matachewan dikes from paleomagnetism and implications for the Precambrian polar wander path, Can. J. Earth Sci. 27, 915–922, 1990.
25 F. Corfu, T.E. Krogh, Y.Y. Kwok and L.S. Jensen, U-Pb zircon geochronology in the southwestern Abitibi greenstone belt, Superior Province, Can. J. Earth Sci. 26, 1747–1763, 1989.
26 C.J. Hale and P. Lloyd, Paleomagnetic analysis of regional and contact strains. Geoscience research grant program: Summary of research 1988–1989, Ont. Geol. Surv. Misc. Pap. 143, 199–209, 1989.
27 F. Corfu and G.M. Stott, U-Pb geochronology of the central Uchi subprovince, NW Ontario (abstract), Geol. Assoc. Can. Program Abstr. A55, 1989.
28 G. Pullaiah, E. Irving, K.L. Buchan and D.J. Dunlop, Magnetization changes caused by burial and uplift, Earth Planet. Sci. Lett. 28, 133–143, 1975.
29 D.J. Dunlop, Paleomagnetism of Archean rocks from northwestern Ontario II: Shelley Lake granite, Quetico Subprovince, Can. J. Earth Sci. 21, 869–879, 1984.
30 G.W. Berger and D. York, $^{40}Ar/^{39}Ar$ dating of multicomponent magnetizations in the Archean Shelley Lake granite, northwestern Ontario, Can. J. Earth Sci. 16. 1933–1941, 1979.
31 P.N. Wilson and W.T. Parry, Mesozoic hydrothermal alteration associated with gold mineralization in the Mercur district, Utah, Geology 18, 866–869, 1990.
32 J.P. Richards and I. McDougall, Geochronology of the Porgera gold deposit, Papua New Guinea: resolving the effects of excess argon on K-Ar and $^{40}Ar/^{39}Ar$ age estimates for magmatism and mineralization, Geochim. Cosmochim. Acta 54, 1397–1415, 1990.
33 L.M. Heaman, A precise U-Pb zircon age for a Hearst dike (abstract), Geol. Assoc. Can. Program Abstr. 11, 77, 1988.
34 P.E. Smith and R.M. Farquhar, U-Th-Pb isotope studies of mafic dikes and gneisses in the KSZ, in: Project Lithoprobe—Kapuskasing Structural Zone Workshop, pp. 187–189, Univ. Toronto, 1989.
35 T.M. Gates and P.M. Hurley, Evaluation of Rb-Sr dating methods applied to Matachewan, Abitibi, Mackenzie and Sudbury dike swarm in Canada, Can. J. Earth Sci. 10, 900–919, 1973.
36 E. Irving and A.J. Naldrett, Paleomagnetism in the Abitibi greenstone belt and Abitibi and Matachewan diabase dikes: evidence of the Archean geomagnetic field, J. Geol. 85, 157–176, 1977.
37 L.D. Schutts and D.J. Dunlop, Proterozoic magnetic overprinting of Archean rocks in the Canadian Superior Province, Nature 291, 642–645, 1981.
38 R.E. Ernst and H.C. Halls, Paleomagnetism of the Hearst dike swarm and implications for the tectonic history of the KSZ, northern Ontario, Can. J. Earth Sci. 21, 1499–1506, 1984.
39 H.C. Halls and E.G. Shaw, Paleomagnetism and orientation of Precambrian dikes, eastern Lake Superior region, and their use in estimates of crustal tilting, Can. J. Earth Sci. 25, 732–743, 1988.
40 M.P. Bates and H.C. Halls, Broad-scale Proterozoic deformation of the central Superior Province revealed by paleomagnetism of the 2.45 Ga Matachewan dyke swarm, Can. J. Earth Sci. 28, 1780–1796, 1991.
41 E. Irving, Paleopoles and paleolatitudes of North America and speculations about displaced terrains, Can. J. Earth Sci. 16, 669–694, 1979.
42 D.J. Dunlop, Paleomagnetism of Archean rocks from northwestern Ontario IV: Burchell Lake granite, Wawa–Shebandowan Subprovince, Can. J. Earth Sci. 21, 1098–1104, 1984.
43 D.J. Dunlop, Paleomagnetism of Archean rocks from northwestern Ontario, V. Poobah Lake alkaline complex, Quetico subprovince, Can. J. Earth Sci. 22, 27–38, 1985.
44 D.T.A. Symons and T.A. Vandall, Paleomagnetic evidence for Proterozoic tectonism in the Kapuskasing Structural Zone, Ontario, J. Geophys. Res. 95, 19199–19211, 1990.
45 T.A. Vandall and D.T.A. Symons, Paleomagnetism of Archean granites and Matachewan dikes in the Wawa subprovince, Ontario: reevaluation of the Archean apparent polar wander path, Can. J. Earth Sci. 27, 1031–1039, 1990.
46 R.G. Gordon, A. Cox and S. O'Hare, Paleomagnetic Euler poles and the apparent polar wander and absolute motion of North America since the Carboniferous, Tectonics 3, 499–537, 1984.
47 E. Irving and J.K. Park, Hairpins and superintervals, Can. J. Earth Sci. 4, 1318–1324, 1972.
48 E. Irving and J.C. McGlynn, Proterozoic magnetostratigra-

phy and the tectonic evolution of Laurentia, Philos. Trans. R. Soc. London A280, 433–468, 1976.

49 D.J. Dunlop, Paleomagnetism of Archean rocks from northwestern Ontario: Wabigoon gabbro, Wabigoon Subprovince, Can. J. Earth Sci. 20, 1805–1817, 1983.

50 D.W. Davis, C.E. Blackburn and T.E. Krogh, Zircon U-Pb ages from the Wabigoon–Manitou Lakes region, Wabigoon Subprovince, northwest Ontario, Can. J. Earth Sci. 19, 254–266, 1982.

51 J.W. Geissman, D.W. Strangway, A.M. Tasillo-Hirt and L.J. Jensen, Paleomagnetism and structural history of the Ghost Range intrusive complex, central Abitibi belt, Ontario: further evidence for the late Archean geomagnetic field of North America, Can. J. Earth Sci. 19, 2085–2099, 1982.

52 V. Costanzo-Alvarez and D.J. Dunlop, Paleomagnetic evidence for post-2.55 Ga tectonic tilting and 1.1 Ga reactivation in the southern Kapuskasing Zone, Ontario, Canada, J. Geophys. Res. 93, 9126–9136, 1988.

53 V. Costanzo-Alvarez, D.J. Dunlop and L.J. Pesonen, Paleomagnetism of alkaline complexes and remagnetization in the Kapuskasing structural zone, Ontario, Canada, J. Geophys. Res. 98, 4063–4079, 1993.

54 D.T.A. Symons, A.W. Quick and M. Stupavsky, Magnetic and paleomagnetic characteristics of the Archean iron formation and host rocks at the Adams mine, Ontario, Ont. Geol. Surv. Misc. Pap. 98, 293–307, 1981.

55 R.H. Ridler and J.H. Foster, Primary paleomagnetic stratigraphy of the Archean Skead pyroclastics, Kirkland Lake area, northeastern Ontario, Geol. Surv. Can. Pap. 77-10, 31 pp., 1981.